Terrain-Adaptive Monocular Depth Estimation for Legged Locomotion

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Abstract

Depth estimation from a single monocular camera remains brittle on the rough, deformable terrain encountered by legged robots. We present a terrain-adaptive depth model that conditions a dense regression head on proprioceptive terrain signals and a learned slip prior. On three outdoor quadruped datasets the method reduces absolute relative error by 18% relative to a strong monocular baseline while running onboard at 24 frames per second.

1 Introduction

Field-deployable perception systems operate under tight power, compute and maintenance constraints. This paper studies the design of such a system for the workshop’s target scenario and reports the engineering choices that mattered in practice. We describe the method, the experimental protocol and the deployment experience.

The contributions are threefold: a compact architecture that meets the onboard budget, a calibration procedure that produces usable confidence estimates, and an evaluation on three real sequences collected with the target platform.

2 Method (continued)

The pipeline is built around a compact backbone with two prediction heads. The first head regresses a dense quantity at half resolution; the second head predicts a per-tile confidence that is used to weight the regression loss. Training uses a mixture of simulated and real sequences, with the simulation configured to match the sensor noise profile of the target platform. We keep the parameter count low enough to fit the onboard accelerator and use a fixed-point export for deployment.

Optimisation follows a two-stage schedule. In the first stage the backbone is trained with photometric supervision only; in the second stage the confidence head is calibrated on a held-out split. We found that the second stage matters most when the platform traverses specular or textureless surfaces, because the confidence signal prevents the regression head from fitting transient artefacts.

Implementation details: all models are trained with a batch size of 32 on four accelerators. The augmentations include random exposure, motion blur and synthetic sensor dropout. We report results from the final checkpoint chosen on the validation split, and we do not use test-time augmentation in any of the reported numbers.

3 Experiments (continued)

Ablations remove one component at a time. Removing the confidence head increases the tail error substantially, while removing the temporal filter affects mostly the short segments with abrupt motion. We also test a reduced-capacity variant to quantify the cost of the onboard constraint. The results show that the full model is the best compromise between accuracy and latency on the target hardware.

The error distributions are heavy-tailed. We therefore report both the mean and the 95th percentile. Across the three sequences the proposed system improves the mean error by 18% and the 95th percentile by 24% relative to the strongest baseline. The improvement is consistent across sequences and does not depend on a single favourable run.

We evaluate on three sequences that differ in terrain, lighting and sensor payload. The primary metric is the error over the full trajectory, with secondary metrics measuring degradation under sensor dropout. All baselines are re-trained on the same data and with the same budget. The comparison therefore isolates the contribution of the proposed components rather than differences in tuning.

4 Deployment (continued)

Power draw stays within the platform envelope. Thermal throttling was observed only during extended direct sunlight and did not affect the control loop. The full configuration files and calibration routines are released with the workshop artefact.

The deployment stack runs on a small companion computer with a fixed power budget. We quantise the backbone and keep the confidence head in full precision because its calibration is sensitive to quantisation. Latency is measured end to end, including preprocessing and the control loop, and the system sustains the required rate with headroom for occasional sensor re-initialisation.

Field trials were conducted over two weeks. Operators reported that the system recovers quickly after a sensor occlusion and that the confidence signal is useful for triggering a safe stop. We logged every intervention and include them in the failure analysis. No hardware failures occurred during the trials.

5 Discussion (continued)

The main lesson is that confidence calibration is as important as raw regression accuracy for field deployment. A model that is slightly less accurate but well calibrated is easier to integrate into a safety envelope, because the downstream planner can react to uncertainty instead of treating every prediction as equally trustworthy.

Limitations remain. The evaluation covers a limited set of terrains and does not include heavy precipitation. The confidence head is calibrated per platform and must be recalibrated after major payload changes. We also did not study long-term drift, which is important for persistent deployments and is left for future work.

Future work will extend the evaluation to snow and mud, integrate a lightweight online calibration routine, and study how the confidence signal can be consumed by a sampling-based planner. We expect the calibration step to be the main engineering cost of transferring the system to a new platform.

6 Related Work (continued)

Our work is closest to hybrid systems that use a learned front end with a fixed backend. The difference is the explicit treatment of terrain and the calibrated confidence head. We do not claim state-of-the-art accuracy on every benchmark; instead we target robust behaviour under the sensor and compute constraints of small field platforms.

A full discussion of related datasets and evaluation protocols is included in the appendix of the extended version of this paper.

Classical approaches rely on geometric consistency and explicit sensor models. Learning-based approaches replace hand-crafted cost functions with differentiable predictors trained on large corpora. Hybrid systems combine a learned predictor with a classical smoother and are often the most practical choice for field robots because they retain the interpretability of the smoother while benefiting from learned features.

7 Conclusion (continued)

The workshop discussion will focus on calibration under distribution shift and on the integration of uncertainty into planning. We welcome feedback from the field robotics community on additional deployment scenarios that would stress the current design.

We presented a practical system for field-deployable perception and evaluated it on three real sequences. The system meets the onboard compute budget and improves both mean and tail error relative to strong baselines. The released configuration and calibration scripts are intended to make the results reproducible on similar platforms.

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